

Jie Ding

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Education

Boston University (BU)

Boston, MA

M.Sc. in Mathematical Finance & Financial Technology

Jan. 2026

Current GPA: 3.91/4.0 (Rank: 2/90)

Selected courses: Real Analysis (PhD), Hypothesis Testing (PhD), Nonparametric and Semiparametric Data Modeling (PhD), Stochastic Calculus, Programming (Python, C++), Computational Methods, Financial Econometrics, Statistical Learning and Deep Learning, Bayesian Modeling.

Shanghai University of Finance and Economics (SUFU),

Shanghai, China

M.Sc. in Applied Statistics

Expected Aug. 2026

Current GPA 3.75/4.0 (Rank: 1/12)

Scholarships: Chinese National Scholarship, First-class Academic Honors for Graduate Students of SUFE

Selected courses: Optimization Methods, Survival Analysis (PhD), Artificial Intelligence, Statistical Learning, Statistical Method and Theory, Game Theory.

Nanjing University of Information Science & Technology (NUIST)

Nanjing, China

B.Sc. in Information and Computation Science

Jun. 2021

Selected courses: Mathematical Analysis I/II/III, Advanced Algebra I/II, Real Function and Functional Analysis, Probability and Mathematical Statistics, Ordinary Differential Equations, Partial Differential Equations, Complex Analysis.

Manuscript in Preparation

Probabilistic Discrete-time Analysis of Stochastic Gradient Algorithms. (Advisor: Dr./Prof. Jonathan H. Huggins).

Research Experience

Boston University, Boston, MA

Oct. 2025-Jan. 2026

Research Assistant (Advisor: Dr./Prof. Mark Kon)

- Built a cleaned NHANES analytic dataset and defined depression as $\text{PHQ-9} \geq 10$. Developed an end-to-end ML pipeline and compared LR/RF/NB/SVM/XGBoost/LightGBM using AUC, F1, sensitivity/specificity.
- Achieved best performance with XGBoost (AUC = 0.73, Accuracy = 0.70, F1 = 0.71) and generated SHAP visualizations to explain key feature contributions.

Boston University, Boston, MA

Jun. 2025-Present

Research Assistant (Advisor: Dr./Prof. Jonathan H. Huggins)

- Extended Wasserstein error analysis from SGD to SGLD, covering both constant and decreasing step-size regimes, with and without momentum.

- Derived generalized error bounds independently, providing self-contained proofs that build on and extend existing results.
- Established stationary Wasserstein-2 error bounds with bias scaling linearly in the step size and inversely in the batch size and inverse temperature, valid for sparse high-dimensional GLMs under standard RSC/RSM conditions.

Shanghai University of Finance and Economics (SUFU), Shanghai, China

Jan. 2024-Present

Participated in the seminar: *Advanced Survival Analysis*

M.Sc. Thesis Project (Advisor: Prof. Miao Han)

Topic: Latent Space-based Network-varying Coefficient Cox Proportional Hazards Model. In progress.

- Proposed a latent space-based network-varying coefficient Cox model that jointly links node-level survival outcomes with network structure.
- Developing an alternating projected gradient algorithm for joint estimation and validated the method via simulation studies and empirical network survival data.
- Derived non-asymptotic error bounds for latent positions, varying coefficients, and the coefficient matrix under suitable regularity conditions.

Course Projects

Advanced Machine Learning (Boston University, MA)

Spring 2024

Topic: Mutual Fund Style Classification from Prospectus

- Built a RAG pipeline to classify mutual-fund styles from prospectuses (461 funds, 3 classes), OpenAI embeddings and Chroma k-NN (k=4), 2,000/200 chunking, GPT-4o-mini, achieved 87.4% accuracy on the original labels.
- Engineered interpretable features, skip-gram Word2Vec and domain knowledge-base similarities, trained SVM/RF/LogReg; SVM performed best on a 64/16/20 stratified split (test accuracy 0.914) using RAG-derived labels.
- Added an evidence-based error audit and a rules-based checker to reconcile conflicting third-party labels and guide label cleanup.

Nonparametric and Semiparametric Data Modelling (Boston University, MA)

Fall 2024

Topic: Normalizing Flows for Density Estimation

- Implemented normalizing flows (NICE, RealNVP) and a classical baseline kernel density estimation (KDE) to model 2D energy-based target densities.
- The method achieved strong performance, successfully capturing fine-grained structural details of the distributions that KDE misses.
- Provided interpretability via the invertible mapping and exact log-likelihood computed from the Jacobian determinant, with visualizations of the learned transformation.

Artificial Intelligence (Shanghai University of Finance and Economics, China)

Spring 2023

- Proposed and implemented RW-GAN, adding residual blocks to both the generator and discriminator and using a Wasserstein objective to mitigate instability and mode collapse in anime-face generation.
- Best FID among baselines, RW-GAN 2042.1 vs R-GAN 2108.7, DCGAN 2183.0, GAN 2659.3, WGAN 2853.4, demonstrating improved generative quality and training stability.

Skills

Programming: R Language, Python, C++, SQL, MATLAB, LaTeX.

Additional Information

Languages: Mandarin (native), English (fluent, IELTS:7.0 (6.0))